Schwarzschild had discovered his spacetime geometry as a solution to Einstein's general relativistic field equation. It was the solution for the exterior of a static star, one that neither implodes nor explodes nor pulsates. However, in 1923 George Birkhoff, a mathematician at Harvard, had proved a remarkable mathematical theorem: Schwarzschild's geometry describes the exterior of any star that is spherical, including not only static stars but also imploding, exploding, and pulsating ones.

For their quick calculation, then, Oppenheimer and Snyder simply assumed that a spherical star, upon exhausting its nuclear fuel, would implode indefinitely, and without probing what happens inside the star, they computed what the imploding star would look like to somebody far away. With ease they inferred that, since the spacetime geometry outside the imploding star is the same as outside any static star, the imploding star would look very much like a sequence of static stars, each one more compact than the previous one.

Now, the external appearance of such static stars had been studied two decades earlier, around 1920. Figure 6.2 reproduces the *embedding diagrams* that we used in Chapter 3 to discuss that appearance. Recall that each embedding diagram depicts the curvature of space inside and near a star. To make the depiction comprehensible, the diagram displays the curvature of only two of the three dimensions of space: the two dimensions on a sheet that lies precisely in the star's equatorial "plane" (left half of the figure). The curvature of space on this sheet is visualized by imagining that we pull the sheet out of the star and out of the physical space in which we and the star live, and move it into a flat (uncurved), fictitious *hyperspace.* In the uncurved hyperspace, the sheet can maintain its curved geometry only by bending downward like a bowl (right half of the figure).